PILSEO PARK

Michigan State University◇ East Lansing MI, U.S. 48824
parkpils@msu.edu◇ Homepage◇ Google Scholar◇ LinkedIn

RESEARCH INTEREST

Face Modeling, Digital Human, Physics Simulation, Neural Rendering, Computer Graphics, 3D/4D Vision

EDUCATION

Ph.D. in Computer Science at Michigan State University 08/2023 – Present
Advisor: Prof. Yiying Tong

M.E. in AI Application at Chung-Ang University 09/2020 – 06/2022
Advisor: Prof. Youngbin Kim
Dissertation: Cross-Modal Semantic Alignments Module for Image-Text Retrieval

B.S. in Computer Science at Oklahoma State University 08/2015 – 05/2019
Cum Laude

EXPERIENCE

Michigan State University – *Research Assistant* 08/2023 – Present

- Collaborated with Qualcomm to develop an efficient Gaussian allocation method for high-fidelity 3D head reconstruction; accepted for publication at *WACV* 2027.
- Developed a generalizable framework for reconstructing animatable 3D Gaussian head avatars from a single image using geometry image and symmetry-guided completion; accepted for publication at *ACCV* 2026.
- Developing a physics-guided framework for human–object reconstruction, focusing on reducing artifacts around hand–object contact and improving computational efficiency.

4by4 Inc. – *AI Engineer* 08/2022 – 06/2023

- Developed a dual-color-space network leveraging global priors from RGB and YCbCr representations for natural and realistic photo retouching; published in *Scientific Reports* (2023).

Chung-Ang University – *Research Assistant* 09/2020 – 06/2022

- Developed a graph-based cross-modal semantic alignment module that explicitly models relationships between image regions and textual tokens for image–text retrieval; published in *Multimedia Tools and Applications* (2024).

PUBLICATION

*: equal contribution

Pilseo Park*, Andrew Hou*, Michel Sarkis, Ning Bi, Yiying Tong, and Xiaoming Liu. "EGGHead: Efficient Generalizable Gaussian Heads from a Single Image," *IEEE/CVF Winter Conference on Applications of Computer Vision **(WACV)***, 2027. **(Round 1)** PDF

Pilseo Park, Fizza Rubab, and Yiying Tong. "Learning Semantic Inpainting for Animatable Gaussian Head Avatars." *Asian Conference on Computer Vision **(ACCV)***, 2026. PDF

Pilseo Park, Ze Zhang, Michel Sarkis, Ning Bi, Xiaoming Liu, and Yiying Tong. "Refined Geometry-guided Head Avatar Reconstruction from Monocular RGB Video," *arXiv*, 2025. PDF

Pilseo Park, Soojin Jang, Yunsung Cho, and Youngbin Kim. "SAM: Cross-Modal Semantic Alignments Module for Image-Text Retrieval," *Multimedia Tools and Applications*, 2024. PDF

Pilseo Park, Heungmin Oh, and Hyuncheol Kim. "Dual-Color Space Network with Global Priors for Photo Retouching," *Scientific Reports*, 2023. PDF

ACADEMIC SERVICES

Teaching Assistant at Michigan State University: CSE 472 Computer Graphics; CSE 471 Media Processing and Multimedia

SKILLS AND INTERESTS

Programming	Python, C/C++, C#, Java, HTML, PHP, OpenGL, MATLAB
Framework	PyTorch, TensorFlow